

Question ID: 6b4707aa

| Assessment | Test | Domain                    | Skill   | Difficulty                               |
|------------|------|---------------------------|---------|------------------------------------------|
| SAT        | Math | Geometry and Trigonometry | Circles | Hard <div></div> <div></div> <div></div> |

Question

Circle A in the  $xy$ -plane has the equation  $(x + 5)^2 + (y - 5)^2 = 25$ . Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the  $xy$ -plane is  $(x + 5)^2 + (y - 5)^2 = k$ , where  $k$  is a constant. What is the value of  $k$ ?

Correct Answer: 100

Rationale

The correct answer is **100**. An equation of a circle in the  $xy$ -plane can be written as  $(x - t)^2 + (y - u)^2 = r^2$ , where the center of the circle is  $(t, u)$ , the radius of the circle is  $r$ , and where  $t$ ,  $u$ , and  $r$  are constants. It's given that the equation of circle A is  $(x + 5)^2 + (y - 5)^2 = 25$ , which is equivalent to  $(x + 5)^2 + (y - 5)^2 = 5^2$ . Therefore, the center of circle A is  $(-5, 5)$  and the radius of circle A is **5**. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is  $(-5, 5)$  and the radius of circle B is **2(5)**, or **10**. Substituting  $-5$  for  $t$ , **5** for  $u$ , and **10** for  $r$  in the equation  $(x - t)^2 + (y - u)^2 = r^2$  yields  $(x + 5)^2 + (y - 5)^2 = 10^2$ , which is equivalent to  $(x + 5)^2 + (y - 5)^2 = 100$ . It follows that the equation of circle B in the  $xy$ -plane is  $(x + 5)^2 + (y - 5)^2 = 100$ . It's also given that the equation defining circle B in the  $xy$ -plane is  $(x + 5)^2 + (y - 5)^2 = k$ , where  $k$  is a constant. Therefore, the value of  $k$  is **100**.